



Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	10	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	5	hr
	Painter	5	hr
Equipment	Excavator	1	hr
	Truck	1	hr
	Generator	1	hr
	Drill	1	hr

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9702/53

October/November 2010

1 hour 15 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

DO **NOT** WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

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1	
2	
Total	

This document consists of **8** printed pages.

- 1 A student wishes to determine the resistivity of aluminium.

The resistivity ρ of a conductor is defined as

$$\rho = \frac{RA}{l}$$

for a conductor of resistance R , cross-sectional area A and length l .

Fig. 1.1 shows the typical dimensions of a strip of aluminium of lengths c , d and t . The resistivity of aluminium is about $10^{-8} \Omega\text{m}$.

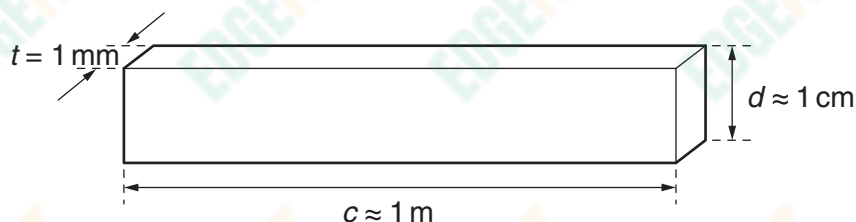


Fig. 1.1 (not to scale)

Design a laboratory experiment to determine the resistivity of aluminium using this strip. The usual apparatus of a school laboratory is available, including a metal cutter.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

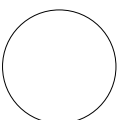
- (a) the procedure to be followed,
- (b) the measurements to be taken,
- (c) the control of variables,
- (d) the analysis of the data,
- (e) the safety precautions to be taken.

[15]

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[illegible]

For Examiner's Use	Defining the problem	Methods of data collection	Method of analysis	Safety considerations	Additional detail



- 2 A student is investigating how the discharge of a capacitor through a resistor depends on the resistance of the resistor.

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The equipment is set up as shown in Fig. 2.1.

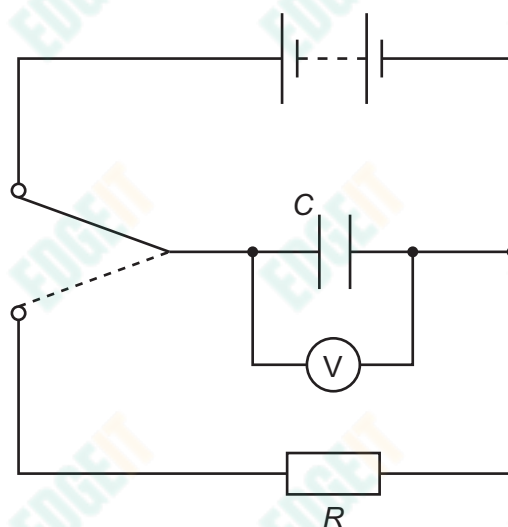


Fig. 2.1

The student charges the capacitor of capacitance C and then discharges it through a resistor of resistance R . After 15.0 s the student records the potential difference V across the capacitor. The student repeats this procedure for different values of R .

Question 2 continues on the next page.

It is suggested that V and R are related by the equation

$$V = V_0 e^{-\frac{t}{CR}}$$

where V_0 is the initial potential difference across the capacitor and t is the time over which the capacitor has discharged.

- (a) A graph is plotted of $\ln V$ on the y -axis against $\frac{1}{R}$ on the x -axis. Express the gradient in terms of C .

gradient = [1]

- (b) Values of R and V for $t = 15.0$ s are given in Fig. 2.2.

$R/\text{k}\Omega$	V/V	$\frac{1}{R}/10^{-6} \Omega^{-1}$	$\ln (V/V)$
6.67	3.6 ± 0.2		
10.0	5.0 ± 0.2		
15.0	6.4 ± 0.2		
20.0	7.2 ± 0.2		
30.0	8.0 ± 0.2		

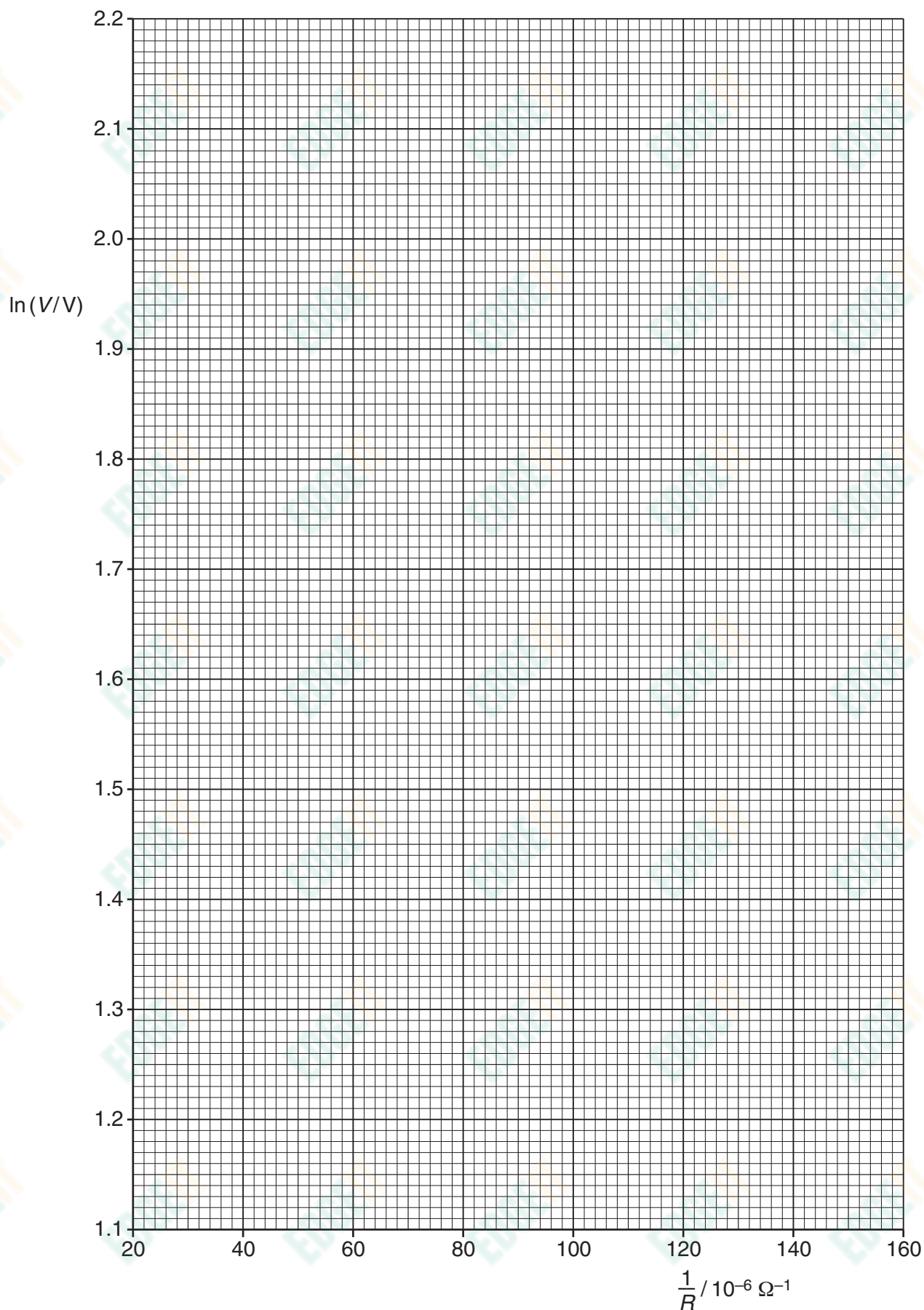
Fig. 2.2

Calculate and record values of $\frac{1}{R}/10^{-6} \Omega^{-1}$ and $\ln (V/V)$ in Fig. 2.2. Include the absolute uncertainties in $\ln (V/V)$. [3]

- (c) (i) Plot a graph of $\ln (V/V)$ against $\frac{1}{R}/10^{-6} \Omega^{-1}$. Include error bars for $\ln (V/V)$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]
- (iii) Determine the gradient of the line of best fit. Include the uncertainty in your answer.

gradient = [2]

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- (d) (i) Using your answer to (c)(iii), determine the value of C . Include an appropriate unit.

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$$C = \dots\dots\dots [2]$$

- (ii) Determine the percentage uncertainty in your value of C .

$$\text{percentage uncertainty} = \dots\dots\dots [1]$$

- (e) Determine the value of R for which the capacitor will discharge to 10% of its original potential difference in 15.0s. Include the absolute uncertainty in your answer.

$$R = \dots\dots\dots \Omega [2]$$

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